Machine Learning Basics

What is Machine Learning?

Machine Learning is a subset of Artificial Intelligence that enables systems to learn and improve from experience without being explicitly programmed. Instead of following explicit instructions, ML systems learn patterns from data.

Types of Machine Learning

1. **Supervised Learning:** Learning with labeled data (e.g., Classification, Regression)
2. **Unsupervised Learning:** Learning patterns from unlabeled data (e.g., Clustering, Dimensionality Reduction)
3. **Reinforcement Learning:** Learning through interaction and rewards

Applications of Machine Learning

- Email spam filtering
- Recommendation systems
- Fraud detection
- Image recognition
- Natural language processing
- Predictive analytics

ML Algorithms & Techniques

Key Algorithms

Regression Algorithms:
• Linear Regression
• Logistic Regression
• Ridge & Lasso Regression

Tree-based Algorithms:
• Decision Trees
• Random Forest
• Gradient Boosting

Clustering Algorithms:
• K-Means
• Hierarchical Clustering
• DBSCAN

Distance & Similarity Metrics

• Euclidean Distance
• Manhattan Distance
• Cosine Similarity
• Hamming Distance

Cross-Validation Techniques

• K-Fold Cross-Validation
• Stratified K-Fold
• Time Series Split

ML Workflow & Best Practices

Steps in Machine Learning Pipeline

1. **Data Collection:** Gather relevant data
2. **Data Preprocessing:** Clean and prepare data
3. **Exploratory Data Analysis (EDA):** Understand data patterns
4. **Feature Engineering:** Create meaningful features
5. **Model Selection:** Choose appropriate algorithm
6. **Model Training:** Fit model to training data
7. **Model Evaluation:** Assess performance
8. **Hyperparameter Tuning:** Optimize model parameters
9. **Deployment:** Put model into production

Performance Metrics

For Regression:
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R-squared (R^2)

For Classification:
- Accuracy
- Precision & Recall
- F1-Score
- ROC-AUC

Common Challenges

- Overfitting: Model memorizes training data
- Underfitting: Model too simple for data
- Class Imbalance: Unequal class distribution
- Feature Scaling: Normalize feature values
- Missing Data: Handle incomplete records